Total No. of Questions : 22 Total No. of Pages : 02

Roll No. ________________

B.Tech IV sem (Main/Back) Exam April-May 2026
Artificial Intelligence & Data Science
4AID4-05 Database Management System
Paper Code : 4E1305
CS, IT, AID, CAI, CCS, CDS, CIT,CSR

Time: 3 Hours **Maximum Marks: 70**

***Attempt all ten questions from Part A,* five questions out of seven questions from Part B and three questions out of five questions from Part C.**

Schematic diagrams must be shown wherever necessary. Any data you feel missing suitably be assumed and stated clearly. Units of quantities used/ calculated must be stated clearly.

Use of following supporting material is permitted during examination. (Mentioned in form No.205)

1 ________________ 2. ________________

Part A (Answer should be given up to 25 words only)

All questions are compulsory

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Q.1 Define DBMS.

Q.2 What is DBA? Mention the functionalities of DBA.

Q.3 Why do we need normalization?

Q.4 Define entity and attribute in ER model.

Q.5 What is weak entity and strong entity?

Q.6 What is selection and projection operation in relational algebra?

Q.7 What is a nested query in SQL? What are the types of nested queries?

Q.8 Define functional dependency with the help of an example.

Q.9 What is a transaction?

Q.10 What is deadlock in DBMS? **10 x 2 = 20**

Part B (Analytical/Problem solving questions)

Attempt any Five questions

Q.1 Explain File System vs DBMS.

Q.2 Describe key constraints and participation constraints in ER model.

Q.3 Explain different types of joins in relational algebra.

Q.4 Explain about the following clauses with example queries.

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(i) Group by

(ii) Order by

(iii) Aggregation functions.

Q.5 Explain Boyce-Codd Normal Form (BCNF).

Q.6 Describe ACID properties of a transaction.

Q.7 Explain conflict serializability with the help of an example

5x 4 = 20

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Part C (Descriptive/Analytical/Problem Solving/Design question)
Attempt any three questions

Q.1 Design an ER diagram for a university database and explain all components.

Q.2 a) Draw and explain the detailed system architecture of DBMS.

b) What are the advantages of DBMS?

Q.3 Explain normalization process with 1NF and 2NF with examples

Q.4 Explain Wait/Die and Wound/Wait Schemes in transaction management.

Q.5 a) Differentiate specialization and generalization.

b) Consider the following database schema to write queries in SQL

Sailor (sid, sname, age, rating)

Boats (bid, bname, bcolor)

Reserves (sid, bid, day)

i) Find the sailors who have reserved a red boat

ii) Find the names of the sailors who have reserved at least two boats

iii) Find the colors of the boats reserved by 'Mohan' **3x10= 3**